

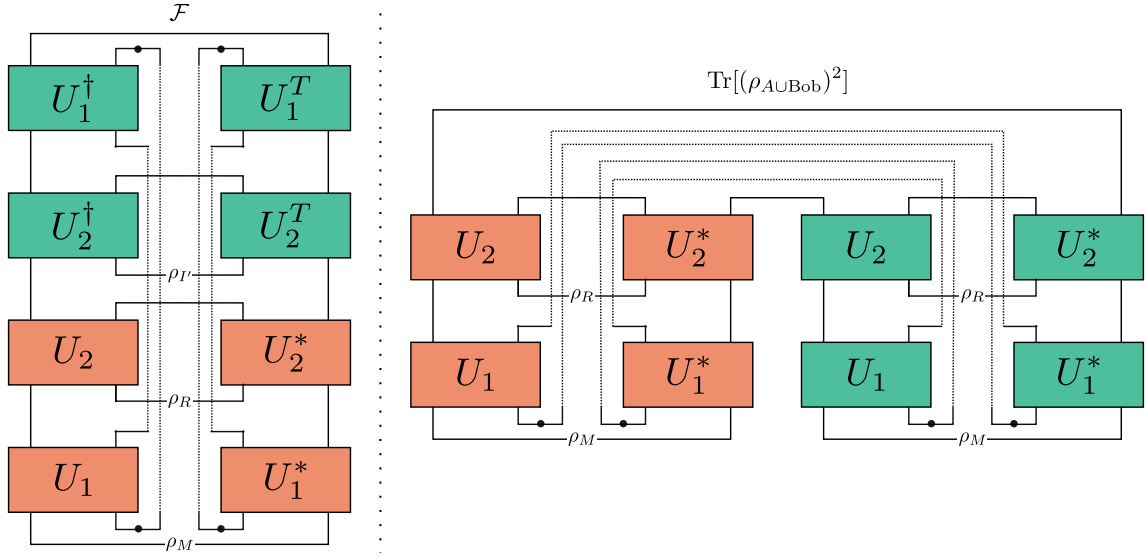


Figure 15: Left: The fidelity of Bob's decoding protocol for a given unitary circuit, U , and initial states: ρ_M , ρ_R and $\rho_{I'}$. Right: The purity of the reduced density matrix $\rho_{A \cup \text{Bob}}$. If ρ_M , $\rho_{I'}$ and ρ_R are all infinite temperature states then these two diagrams are equivalent down to a numerical factor.

B Derivation of Eq. (11)

The fidelity of the decoder for Bob's protocol is related to the purity of the reduced density matrix $\rho^{A \cup \text{Bob}}$ prior to decoding via

$$\mathcal{F} = d_M^{-1} \text{Tr}(\rho_{A \cup \text{Bob}}^2), \quad (18)$$

where d_M is the Hilbert space dimension of the medium; we show this diagrammatically in Figure 15. By manipulating the fidelity diagram the decoder becomes a replica in the purity diagram. These two diagrams correspond when ρ_M , $\rho_{I'}$ and ρ_R are infinite temperature states. Infinite temperature states in the fidelity are related to trace operations in the purity (and *vice versa*) via

$$\rho_B^\infty \leftrightarrow d_B^{-1} \text{Tr}_B[\cdot], \quad (19)$$

where d_B is the Hilbert space dimension of B . These dimensional factors mostly cancel in our case except for d_M .

Now, the entanglement spectrum for Clifford circuits is flat[]; this results in an equivalence between Renyi entropies, $S_n(\rho_C)$, defined by

$$S_n(\rho_C) = \frac{1}{n-1} \log_2 \text{Tr}[\rho_C^n]. \quad (20)$$

This combined with the relation $S(\rho_C) = \lim_{n \rightarrow 1} S_n(\rho_C)$ allows us to make the following substitutions:

$$\begin{aligned} I(A \rightarrow \text{Bob}) &= S_2(\rho_{\text{Bob}}) - S_2(\rho_{A \cup \text{Bob}}) \\ &= -\log_2 \text{Tr}[\rho_{\text{Bob}}^2] + \log_2 \text{Tr}[\rho_{A \cup \text{Bob}}^2] \\ &= 1 + \log_2 \mathcal{F}, \end{aligned} \quad (21)$$

where to arrive at the last line we used $S_2(\rho_{\text{Bob}}) = -\log_2(d_M d_I) = 1 - \log_2(d_M)$.